

Cortico-Cerebellar model

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Cortex hallucination of sensory input

$$\hat{\mathbf{I}} = \mathbf{W}_{gen} \cdot \boldsymbol{\mu}_t \quad (1)$$

- ▶ $\hat{\mathbf{I}}$, is the *hallucination* of equal dimension to $\mathbf{I}$ *actual input*
- ▶ $\mathbf{W}_{gen}$ is the generative matrix than makes the lower-dimension projection, intialized random and fixed $\mathcal{N}(0, 1/\sqrt{\dim \boldsymbol{\mu}})$
- ▶ $\boldsymbol{\mu}_t$ is the *cortical state* at time t

Cortical prediction error

$$\epsilon = \Pi \odot (\mathbf{I}_t - \hat{\mathbf{I}}_t) \quad (2)$$

- ▶ The cortex makes a *hallucination* which acts as active inference of the world, then the participant makes an actual choice generating the sensory vector $\mathbf{I}_t$
- ▶ To build Π as the precision weighting, we choose the most simplest one. Static anatomical precision, let $\Pi_{raw,i} = 1/\sigma(\mathbf{I}_i)$ be the raw precision weighting vector for input i with variance $\sigma(\mathbf{I}_i)$, then we generate $\Pi = \Pi_{raw}/\bar{\Pi}_{raw}$ to be the normalized precision weighting
- ▶ Element-wise multiplication means we build prediction error ϵ as the precision weighted prediction error

Propagation of cortical state in to the Cerebellum

$$\mathbf{G}_t = \mathbf{W}_{ach1} \cdot \boldsymbol{\mu}_t \quad (3)$$

- ▶ $\mathbf{G}_t$ is the Granule Layer state at time t , defined as a Marr-Albus-Ito random sparse projection
- ▶ We modeled this as an Achlioptas random sparse projection (896×32), where each weight $w_{i,j}$ is assigned with probability p : -1 or 1 , and 0 with $1 - 2p$, setting sparsity $(1 - 2p)$ to 90% zeros

Cerebellum memory or eligibility trace

When the cortex makes a prediction and initiates an action, there is some delay for this feedback to arrive. How do we solve the problem of pairing some particular $\mathbf{G}_t$ with a given feedback. We need to leave an *eligibility trace*

$$\mathbf{Z}_t = (1 - \beta_{gran}) \cdot \mathbf{Z}_{t-1} + \alpha_{gran} \cdot \mathbf{G}_t \quad (4)$$

- ▶ Where $\mathbf{G}_t$ is the instantaneous firing of the Granule Layer
- ▶ α_{gran} is the parameter that controls how aggressively the layer updates
- ▶ $(1 - \beta_{gran})$ is the *leaky bucket*, that decays the trace on every trial

Cerebellum learning

To learn the Cerebellum projects the 6-Dimensional precision weighted error ϵ to a single dimension (relating this to the Inferior Olive)

$$|\bar{\epsilon}| = \frac{1}{d} \sum_{i=1}^d |\epsilon_{t,i}^-| \quad (5)$$

With that scalar error we update the Granule Layer trace in its Purkinje Cell representation via anti-Hebbian learning

$$W_{purk,t} = W_{purk,t-1} - \kappa(|\bar{\epsilon}| \cdot \mathbf{Z}_t) \quad (6)$$

Cerebellum compression

$$\mathbf{D}_t = \mathbf{W}_{ach2}(\mathbf{G}_t \odot \mathbf{W}_{purk,t}) \quad (7)$$

- ▶ $\mathbf{D}_t \in \mathbb{R}^{362}$ is the Cerebellum forward pass
- ▶ $\mathbf{G}_t \odot \mathbf{W}_{purk,t}$ is the element-wise multiplication between the Granule Layer current state and the LTD Purkinje layer we previously constructed
- ▶ Finally with $\mathbf{W}_{purk,t}$ we perform a random sparse matrix compression into the 362 dimension of the cerebellar forward pass

The loop-back: thalamo-cortical pass

$$\mu_{t+1} = \mu_t + \alpha_c [(\mathbf{W}_{gen}^T \cdot \epsilon_t) - (\lambda_c \cdot ITI_t \cdot \mu_t) + \beta_{thal}(\mathbf{W}_{thal} \cdot \mathbf{D}_t)] \quad (8)$$

- ▶ This is the Cortical trace update, $\mathbf{W}_{gen}^T \cdot \epsilon_t$ projects the error back into the cortical dimension $6 \rightarrow 32$, and we perform gradient descent over a time-decaying trace $\lambda_c \cdot ITI_t \cdot \mu_t$
- ▶ $\mathbf{W}_{thal}$ is the connectivity from cerebellum to cortex $362 \rightarrow 32$, so $\mathbf{W}_{thal} \cdot \mathbf{D}_t$ is the cerebellum-cortex forward pass, which via β_{thal} pushes the cortex into the non-linear projection
- ▶ Finally, the trace is updated via learning rate α_c

Converting a trace into action and reaction-time

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